

LandWeb Code and Module Changes

What changed from v2 to v3, and why

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⚠ Warning

Draft report: pending human review

Portions of this document were drafted with the help of generative AI (large language models) and have not yet been fully reviewed by the co-authors. Content may be incomplete, inaccurate, or misattributed. This notice will be removed once a complete review has been carried out.

1 Overview

LandWeb v3 is a substantial rebuild of the modelling infrastructure rather than a change of ecological content. The simulation science — the **LandR** Biomass vegetation modules (McIntire et al. 2025) and the LandMine fire model — is retained, but the way those modules are **orchestrated, cached, and dispatched** has been rewritten, and the input data modernized (see the study-area vegetation and fire-model-inputs reports). This report reconstructs the code and module changes across the v2 (2020–2023) to v3 (2024–2026) window.

NEWS files were not maintained across this window, so the change history here is reconstructed from each module’s and package’s **git log**. The per-component commit counts, last-update dates, and currently-pinned revisions are generated automatically (see the Appendix) and are therefore current as of this render.

2 From manual orchestration to a targets pipeline

The defining v3 change is orchestration. In v2, a chain of hand-run scripts (00-main.R ... 03-main-sim.R) drove `simInit()/spades()` calls in sequence, persisting whole `simList` objects between stages and relying on the **reproducible** (McIntire and Chubaty 2024) cache for invalidation. In v3, a **targets pipeline** owns orchestration and caching; the SpaDES modules (Chubaty and McIntire 2025) remain the simulation engine but are wrapped as targets.

Concern	v2 (manual SpaDES)	v3 (targets pipeline)
Orchestration	hand-run 00–03 scripts	declarative <code>_targets.R</code> DAG
Stage hand-off	serialized <code>simList</code>	components extracted, passed as plain objects
Caching	reproducible cache (Postgres/SQLite)	<code>targets;</code> <code>reproducible.useCache = FALSE</code> everywhere
Persistence	<code>saveSimList()</code> / <code>loadSimList()</code>	per-object <code>format = "file"</code> targets (qs2, spatial)

Concern	v2 (manual SpaDES)	v3 (targets pipeline)
Reps / scaling	~15 reps/area, one machine	many reps; multi-node via <code>crew.ssh</code>
Post-processing	3 summary modules	consolidated into <code>NRV_summary + nrvtools</code>

The motivation was reliability, not preference: the `reproducible` cache and `saveSimList()` were fragile with environment references and `terra` pointers, and upstream framework churn (below) kept breaking long-running configurations. Routing caching and orchestration through `targets` — which invalidates by content and rebuilds only what changed — removed a large, recurring source of failure and made the multi-machine rep-scaling tractable. This is the PERFICT-style continuous, tested, reusable workflow argued for in McIntire et al. (2022) and Barros et al. (2023), realized for LandWeb.

3 SpaDES module changes

Each module is documented below with a high-level summary followed by grouped detail, reconstructed from its git history; commit magnitudes and current pins are in the Appendix. The recurring cross-cutting themes are **SCANFI data adoption** (replacing kNN/Beaudoin and NTEMS/Pickell inputs), the **qs2** serialization switch, **terra** migration, and adoption of **FOR-CAST forks** of the `Biomass_*` modules (pinned, self-backported). Modules are ordered by pipeline stage: data preparation, simulation, post-processing, then the retiring modules.

3.1 LandWeb_preamble

The study-area/data-prep setup underwent a large Phase-0 rewrite onto `terra/tidyterra` for the `targets` pipeline, swapped in SCANFI as the current-conditions data source (with hand-rolled `prepInputs/Drive` workarounds), reworked the species set and labelling, and added Spray Lake + C5 (Douglas-fir) landbases. As of mid-2026 Phase-0 runs end-to-end for SprayLake; several data steps (30 m CC maps, NTEMS age adjustment, `mergeSlivers` migration) remain TODO.

3.1.1 Phase-0 terra/targets port

- ~938-line rewrite of `LandWeb_preamble.R` porting the module to `terra/tidyterra` for the `targets` pipeline; Phase-0 runs for SprayLake (c00e79a, 2026-06).
- Introduced `studyAreaReporting/rasterToMatchReporting` outputs and rebuilt geoprocessing on `SpatVector/SpatRaster` (`terra::vect`, `terra::project`, `terra::expanse` in place of `sf::st_area`), touching all per-FMA R/*.`R` helpers.

- Removed the `map` package dependency and moved plotting functions to `LandWebUtils`; adopted the native pipe (`f0e33c2/a263b57`, 2025-08).

3.1.2 SCANFI integration and prepInputs workarounds

- Reworked current-conditions data prep around SCANFI, expanding required packages (`dplyr`, `geodata`, `httr`, `ngeo`, etc.) (`5e0df33/bd3b1d2`, 2025-09).
- Bypassed `LandR::prepInputs_SCANFI_LCC_FAO` (its FAO step downloaded/processed the ~840 MB Canada-wide raster, over 50 min) with a windowed download-then-crop of SCANFI LCC + FAO; flagged as a future `prepInputs_SCANFI_LCC_FAO_fast` `LandWebUtils` helper (`c00e79a`, 2026-06).
- Dropped the unreliable SCANFI `Popu_gra` species layer and switched tree classes to integer (`e5e5215/4a57674`, 2025-10); `LandTypeCC` left all-NA to defer to SCANFI LCC pending 30 m CC maps.

3.1.3 Species relabelling and expanded species set

- Simulate more tree species with reworked species params (`bd3b1d2`, 2025-09).
- Relabelled western redcedar/western hemlock (`THUJ_PLI/TSUG_HET`) to `Abie_spp` (fir), treating them as SCANFI over-attribution in AB (`a101a9f`, 2026-06).
- Labelled *Pinus contorta* varieties (shore vs lodgepole) and dropped the vestigial `PINU_CON` in `InitSpecies` (`8033234`, 2026-06); added generic names and leading for Douglas-fir (`a2b9ca6`, 2024-09).

3.1.4 Google Drive service-account auth and large-file downloads

- Fetch the v10 LTHFC map via `workflowtools::drive_download_once` (googledrive direct, service-account auth) to bypass reproducible's Drive path that lost SA auth (reproducible #447) (`c00e79a`, 2026-06).
- Pull the ~5.5 GB SCANFI 2020 stand-age raster through the authenticated googledrive API because a fresh reproducible download returned an unauthenticated sign-in HTML page; raw SCANFI median age used for now, skipping NTEMS fire/harvest + kNN adjustment (TODO).
- Removed the interactive `browser()` debug call left in the SCANFI rework during the Phase-0 port.

3.1.5 Study-area / FMA rework (Spray Lake + C5, LTHFC)

- Added the Spray Lake + C5 landbase and its `R/SprayLake.R` builder (`f94a650`, 2024-08), later giving Spray Lakes custom LTHFCs and including Douglas-fir (`c260a06`, 2024-09).

- `studyAreaName` checks updated to allow v3 SprayLake runs; adopted `ltfc_sls_v3`; multiple C5-boundary fixes (2024-09/10).

3.1.6 Object/naming restructuring and reporting polygons

- Renamed core objects for consistency: `rasterToMatchLarge` to `rasterToMatch_biomassParam`, `StudyAreaLandWeb` to `studyAreaLandWeb`, `rstFlammable` to `flammableMap` (2025-10; 65dc365, 2026-07).
- Reporting-polygon builder work: national ecozones + ecoregions reporting polys for AB (1eb353c, 2024-05), MB and v8c LTHFC polygons, sliver-merge/`write_sf` cleanup and `maskTo`-based `postProcess` (2023-09/2024-08).

3.2 Biomass_speciesData

Migrated from `raster/sp` to `terra/sf`, then in v3's defining change switched its default tree-species cover source from NFI kNN to the 30 m SCANFI product (with kNN and several proprietary inventories retained as selectable `types`). Alongside this it gained more robust species inclusion/exclusion, CRS handling, caching tags, and modernized plotting/metadata. Runs on the FOR-CAST fork line (merged to main 2026-06-29).

3.2.1 SCANFI species-layer adoption (replacing kNN)

- Changed the default `types` parameter from “KNN” to “SCANFI”, the 30 m Spatialized Canadian National Forest Inventory product, driven via `LandR::prepSpeciesLayers_SCANFI` (2e3fdc0, 2025-09).
- Expanded selectable layer sources documented in metadata: CASFRI, ForestInventory, KNN, MBFRI, NTEMS, ONFRI, Pickell (most non-kNN requiring proprietary access); added references for KNN, SCANFI, NTEMS (2025-09).
- Handled SCANFI's 30 m native resolution by aggregating to an RTM resolution (240 m) and using SCANFI's projected CRS, since LandR requires projected rasters for dispersal (2025-09).

3.2.2 terra/sf spatial migration and CRS alignment

- Switched spatial input/output object classes to `terra` and `sf`; adopted `postProcessTo/projectTo/cropTo/maskTo` for `rawBiomassMap` and layer preparation instead of `spTransform/raster::mask` (2023).
- Late CRS-robustness fix: use `.compareCRS` instead of `terra::compareGeom` (study area may not be a `terra` object); require RTM as input and standardize on metres (2025).

3.2.3 Species inclusion/exclusion and layer integrity

- Filter out species layers with no data or cover below `coverThresh` after overlaying, with messaging on removed/retained species (2023-02).
- Ensured correct species names are assigned to raster layers, including a fix for missing layer names by falling back to species names (2023-03 / 2024-07).

3.2.4 Caching, plotting and metadata

- Added `.studyAreaName`, `dataYear` to `speciesLayers` cache tags for correct keying across years/areas; corrected `useCache` being character not logical (2023-24).
- Replaced `plotVTM` with the `Plots` function; piped to `Cache()`, removed `pryr/figurePath()`, adopted `sppHarmonize` (2023-2026).
- Renamed the study-area-layers object `SAL` to `studyArea_biomassParam`; default `sppEquivCol` set to “LandR”; repeated LandR/SpaDES.core minimum-version bumps (2025).

3.3 Biomass_borealDataPrep

Ingests national forest-inventory rasters (land cover, stand age, species cover, biomass) and fits the statistical models (biomass vs. age/cover, `maxB`, `maxANPP`, species establishment, per ecoregion x land-cover group) that parameterize `Biomass_core`. The headline v3 changes were a full terra migration, moving the input data pipeline from CASFRI/kNN/NTEMS to SCANFI, new longevity-vs-age and young-biomass corrections, and hardening of the nested `Biomass_core` run; it remains the heaviest and most actively developed module (currently v1.5.12).

3.3.1 Input data: terra migration and SCANFI/NTEMS

- Ported all spatial I/O from `sp/raster` to terra `SpatRaster/SpatVector`, replacing `compareRaster/spTransform/postProcess` with `.compareRas/projectTo/postProcessTo` (2023).
- Switched land cover from static LCC to NTEMS LCC selectable by `dataYear` (2024), then made SCANFI 2020 the default source for age, LCC, species cover and raw biomass, adding a `dataSource` parameter (default “SCANFI”) and `dataYear` (default 2020) (2025, #95).
- Added NTEMS raw-biomass ingestion (with an `archive` dependency to `unzip`), documenting NTEMS/SCANFI provenance in metadata and the bibliography (2025).

3.3.2 Land cover and disturbance handling

- Reworked the “adjust LCC” logic that reassigns non-forest/disturbed pixels: NTEMS layers to backfill missing LCCs, treed wetland (class 81) added to forested classes, cleaner handling of empty pixels / `pixelsToRm` / `pixelFateDT` (2025); later migrated the LCC-disturbance backfill from NTEMS to SCANFI (2025).
- Parameterized forested classes via `forestedLCCClasses`, and added a `landis` mode that collapses all forested classes to one so `ecoregionGroup` is defined by ecoregion only (for LANDIS-II Biomass Succession), default `FALSE` (2026).

3.3.3 Biomass and age parameterization

- Added a longevity-vs-age adjustment reconciling species ages against LandR longevity, with a diagnostic figure (`ageAdjustmentPlot`) (2025).
- Fixed young-biomass estimation (spinup): allow `minCohortBiomass` = 0, avoid negative biomass at low cover via `pmax`, relaxed the young-B assertion to tolerate NAs (2024-2025).
- Handled multiple-to-1 mappings in `sppEquiv` (requiring updated LandR); default `sppEquivCol` set to “LandR” (2025).

3.3.4 Serialization and caching

- Migrated serialization to `qs2` (from `qs`) in the module and `updateYoungBiomasses` (2026).
- Tightened caching: explicit `.functionName` tags on many `Cache()` calls, piped to `Cache()`, reworked `speciesEcoregionStack/modelBiomass` caching to fix incorrect cache triggers (2023-2025).

3.3.5 Nested `Biomass_core` handling and fork pinning (*FOR-CAST fork*)

- Moved nested-module code under a `submodules/` subdir (2023-2024).
- Pinned `Biomass_core`: dropped the runtime `getModule("Biomass_core@development")` fetch and `SpaDES.project` dependency, requiring a pinned `Biomass_core` ($\geq 1.3.10$) symlinked into `submodules/` (2026); re-assert `spades.loadReqdPkgs=TRUE` before the inner run and hardened the symlink against stale dirs (2026).

3.3.6 Framework robustness and metadata

- Fixed a cluster of fire-behavior/perimeter bugs and multiple LandR version bumps; resolved “freq length zero” / empty-data.table errors; various data.table warnings and reprojection/whole-number-pixel assertions (2024-2025).

- Decoupled the module so it need not run alongside the whole workflow (standalone runs; allow `studyArea` with `NROW` over 1) (2023-2026).
- Migrated plotting to `Plots` + `terra`, converted messaging from `crayon` to `cli`, and repeatedly refreshed the auto-rendered manual Rmd + `render-module-rmd.yaml` GitHub Action (2024-2026).

3.4 Biomass_speciesParameters

Substantially reworked in v3: the GAMM-based growth-curve machinery was replaced with lighter non-linear growth curves fit to a modernized PSPclean data pipeline, the big-data backend moved from the deprecated `disk.frame` to `arrow`, and a new opt-in LANDIS mode exposes fitted curves as LANDIS-II Biomass Succession inputs. Runs on the FOR-CAST fork line (merged to main 2026-06-29), currently v3.0.1.

3.4.1 PSP data / PSPclean pipeline

- Removed ~200 lines of in-module PSP input handling in favour of the external PSPclean package; added New Brunswick data (2023-12).
- Repeated PSPclean version bumps and adaptation to its evolving output; late-2025 rework for a “new data format” and trimming of extraneous columns (2025-12).
- Added guards for insufficient/empty PSP data (warn on too few records, catch empty `PSP_sa`, fix age-1 stands); more efficient sampling of `standAge` from the factorial (2024-2025).

3.4.2 Growth-curve fitting

- Major “deGAMM” rewrite: dropped GAMMs (`mgcv`) for non-linear growth curves, renaming objects/args throughout `anPPfunctions.R`; removed `speciesGAMMs` from the `simList` with RAM-management cleanup (2023-11/12).
- Improved detection/removal of bad/non-converging fits and no-data cases; “HeightFix” and quantile-based `deltaDiff` trait adjustments (2024-2025).

3.4.3 LANDIS-mode outputs

- Added a `landis` parameter (default `FALSE`) exposing fitted LANDIS-version curves as new output `speciesGrowthCurvesLandis` for LANDIS-II Biomass Succession (2026-06).
- Added companion output `speciesGrowthCurvesPSP` (source PSP observations behind the fits) for plotting fitted curves vs PSP points; both outputs always created as empty `data.tables` when `landis = FALSE` to preserve the `createsOutput` contract (2026-06).

3.4.4 Big-data backend and caching

- Migrated off deprecated `disk.frame` to `arrow`: `cohortDataFactorial/speciesTableFactorial` passed as `arrow` dataset paths (produced by `Biomass_speciesFactorial >= 1.0.0`), using `dplyr` syntax and `fs` for factorial file paths (2025-10).
- Cached `modifySpeciesTable` with `omitArgs/cacheExtra` to avoid oversized/long caches (2023-08, 2025-11).

3.4.5 Framework, metadata and API

- Replaced deprecated `crayon` with `cli`; adopted native pipe and piping to `Cache()` (2025-10).
- Object/API cleanups: default `sppEquivCol` to “LandR”; removed `pixelGroup` as a joining column; added modules to `loadOrder`; dropped unused `magrittr` dependency (2025-2026).

3.5 Biomass_speciesFactorial

A smaller module retooled in v3 for the targets pipeline: `registerOutputs` wiring, `qs2/arrow` storage backends, decoupled (no-longer-always-run) execution, and a June-2026 push to pin and correctly resource the nested `Biomass_core` run. Runs on the FOR-CAST fork line (merged to main 2026-06-29), currently v1.0.4.

3.5.1 registerOutputs wiring for targets

- Introduced `registerOutputs` to expose factorial products to the pipeline (2026-01); iterated through breakage after an earlier “can’t registerOutputs until it’s fixed” hold; final fix required strict character-type arguments (2026-06).

3.5.2 qs2 serialization and arrow/fs storage

- Migrated serialization from `qs` to `qs2` in code and `Rmd`; replaced `disk.frame` with `arrow` for the factorial tables; moved factorial-table file paths to `fs` (2025-10/2026-01).

3.5.3 Nested Biomass_core pinning and reqdPkgs (FOR-CAST fork)

- Pinned the nested `Biomass_core` (dropped `SpaDES.project/getModule`, symlinking the project’s module instead), then hardened that symlink against stale directories (2026-06).
- Fixed nested-run dependencies: declared `pemisc` for `factorValues2`, then converged on re-asserting `loadReqdPkgs` rather than over-broad `reqdPkgs` (2026-06).

3.5.4 Decoupled execution and cohort parameters

- Stopped forcing users to run all modules together (2024-04); moved `runExperiment/readExperimentFiles` into `init`; switched factorial runs to `times/end` instead of hardcoded 0 to 100 (2025-12).
- Added `minCohortBiomass`, then `initialB/minCohortB` params with NA-`initialB` guards, passing `initialB` through the factorial (2025-2026).

3.6 Biomass_core

The most heavily developed of the simulation modules: a new diagnostic/transition plotting suite, ggplot2 v4 and cli migration, a SCANFI-driven v2.0.0 species-layer refactor, restored LandR.CS climate-sensitive support, and a long tail of cohort-correctness and dependency modernisation. Active on the adopted FOR-CAST fork line (main merged 2026-06-29), currently v2.0.2.

3.6.1 Diagnostic and transition plotting

- Added a species transition-plot subsystem: new `.plotTransitionField/.plotTransitionTimes` params, a `plotTransitions` event, and `vegTransitions()/plotVegTransitions()` driven by `studyAreaReporting` zones (217ae38, 2026-01).
- Added a relative-biomass plot and a large plotting revamp (subtitles, presentation, new plots); switched default `.plots` output to png (2025).

3.6.2 ggplot2 v4 and cli migration

- Fixed summary plotting for ggplot2 v4.0 (2025-10); replaced deprecated `crayon` with `cli` for console messaging (2025-10).

3.6.3 SCANFI species-layers support (v2.0.0)

- Bumped module to v2.0.0 for SCANFI; reworked `sim$speciesLayers` sourcing and added SCANFI resolution/projection checks when building the RTM; documented `dataSource/dataYear` and added SCANFI/NTEMS citations (2025-10).

3.6.4 LandR.CS climate-sensitive support

- Restored/fixed integration with LandR.CS (climate-sensitive growth/mortality), pinning its dev version rather than master (2026-02).

3.6.5 Cohort and biomass correctness fixes

- Multiple `minCohortBiomass` fixes; `latlon + sf` handling; universal seed-dispersal bugfix; identical-cohort edge case (2024-2025).
- Standardised on `LandR::cohortDefinitionCols()`, ensuring cohorts are not defined by `ecoregionGroup` or `B` (2024).

3.6.6 Dependency and cache/API modernisation

- Updated `Cache` calls to the pipe form; simplified `inputObjects` (removed `SAL`); switched to stable package versions where possible; filtered `sppEquiv/sppColorVect` to simulated species (2024-2025).

3.7 Biomass_regeneration

The lightest-touch module: essentially maintenance in the v3 window, focused on aligning cohort definitions with `Biomass_core/LandR`, finishing the terra migration, and metadata/manual housekeeping. Stable; currently v1.0.1.

3.7.1 LandR cohort-definition alignment

- Added `loadOrder = list(after = "Biomass_core")` (2024-01); adopted the updated `LandR::cohortDefinitionCols()`, ensuring cohorts are not defined by `ecoregionGroup` or `B` (2024).

3.7.2 terra migration and maintenance

- Removed `getValues()` in favour of `as.vector(rstObj[])` for terra compatibility (2023-09).
- Switched to `inputPath` over `dataPath`; metadata cleanup; reformatted the manual `Rmd` and refreshed the `render-module-rmd` GHA workflow (2024-2026).

3.8 LandMine

The fire model was ported off `raster/quickPlot` onto `terra/ggplot2` and its spread parameters re-fit with `DEoptim` for 240 m pixels, while the reburn logic, “burny” discontinuous-fuel handling, and reproducible/SpaDES plumbing were hardened. As of mid-2026 the terra rewrite runs and object names are aligned to the `scfm/fireSense` convention; a few pieces (`registerOutputs`) are freshly re-enabled.

3.8.1 terra migration (away from raster/quickPlot)

- Large 2025-10 rewrite converting all I/O metadata from `Raster` to `SpatRaster`, dropping `quickPlot/fasterize/magrittr` for `terra/tidyterra`; replaced `compareRaster` with `terra::compareGeom` and `raster()/freq` with `terra::rast/terra::freq` (aea149b, 2025-10).
- Rebuilt all plots as `ggplot2 + tidyterra::geom_spatraster/geom_spatvector` (fire-return-interval, flammability, cumulative-burn) writing PNGs, superseding the old `Plot()` path (2025-10).

3.8.2 Fire-spread re-optimization (DEoptim)

- Full LandMine optimization rerun with refreshed fire-size diagnostic figures and updated `data/LandMine_DEoptim_params.csv`; adopted the latest optim values tuned for 240 m pixels (2025-10).

3.8.3 Reburn logic and “burny” robustness

- Two-phase reburn scheme (reignite undersized fires up to `maxReburns`, then start new fires sized to the shortfall); added the `maxReburns` parameter, later per-phase values (2023-09).
- Added `R0Stype` (“burny”/“equal”/“log”/“default”) so areas with discontinuous fuels can still spread fire; fixes for Douglas-fir study areas and areas adjacent to polygons lacking an LTHFC (2023-2024).

3.8.4 Reproducible / SpaDES modernization

- `reproducible` arg rename `cacheRepo` to `cachePath` in `EstimateTruncPareto` (2026-06); corrected `registerOutputs()` usage, temporarily disabled then re-enabled (2025-10 to 2026-07).
- Moved burn/optim/helper functions out of the module into `LandWebUtils` (2023-09); standardized fire object names to the `scfm/fireSense` convention (`rstFlammable` to `flammableMap`, `rstCurrentBurnCumulative` to `burnMap`) (2026-06/07).

3.8.5 Study-area consistency, diagnostics and manual

- Resolved `studyAreaName` inconsistencies; added an LTHFC/FRI summary plot and table; fixed a `rstCurrentBurnCumulative` zero-fill bug and consolidated plot devices to a single managed device (2023).

3.9 NRV_summary

New-for-v3 consolidated post-processing module (first drafted Oct 2022 from a BC NRV script, matured 2024-2026); it computes natural-range-of-variation summaries (leading vegetation, age classes, large patches, seral stages) over reporting polygons and **supersedes the older LandWeb_summary, timeSinceFire, and LandWeb_output modules**. Functional in both single and multi modes.

3.9.1 Module consolidation and single/multi mode

- Converted the standalone BC NRV post-processing script into a SpaDES module (2022-10).
- Major single/multi refactor absorbing LandWeb_output (and WIP LandWeb_summary): single mode saves the inputs multi mode needs as its own outputs to avoid re-loading whole simLists; adds registerOutputs(), switches to qs2, drops the map object, uses terra vectors and tidyterra+ggplot2 for plotting (835b364, 2026-02).
- Added a postprocessEvent parameter to select which post-processing events run, enabling province-specific use e.g. BC vs ON (2024-05).

3.9.2 reportingPolygons named-list (replacing the map/ml god-object)

- Removed the ml (map list) output object entirely (2026-03).
- Stripped all sim\$reportingPolygons@metadata “map vestige” lookups across landscapeMetrics/patchMetrics/patchMetricsSeralBC/plotFun, replacing md[layerName == p] indexing with per-polygon named-list access (rptPoly[["ID"]], "NAME") (2026-03).
- Switched per-polygon column selection to the list element’s field attribute instead of the old useCol attribute (2026-02).

3.9.3 start(sim)-relative times and summaryPeriod

- Made summaryPeriod and timeSeriesTimes default to start(sim) + their offsets, keyed the scheduled events off start(sim), and hardened init-time range validation against start(sim)/end(sim) (2026-02/03).

3.9.4 standAge / vegType wiring

- Moved vegTypeMap/standAgeMap generation off sim\$ onto mod\$ (module-local), writing rasters to disk and registering them rather than carrying them in the simList; declared both as explicit createsOutput objects (2026-03).

- Relocated object-presence/consistency checks out of `.inputObjects` into `init` so validation runs against the assembled `simList` (2026-03).

3.9.5 Plotting and seral stages via `nrvttools`

- Depends on `nrvttools` for seral-stage map creation (bumped to pick up fixes); added/corrected the `sieveThresh` control (2024-06).
- Added violin plots, multi-page patch-metric plots, seral-stage patch figures; masked non-flammable pixels in time-since-fire maps that otherwise skewed ages very old (2023-2024).

3.10 burnSummaries

Fire/burn post-processing module extracted from the main LandWeb repo (Sep 2023) for v3; it derives burn maps, fire-size distributions, cumulative (mean-annual) burn history, and time-since-fire from simulation outputs, comparing simulated to observed/expected fire regimes. Functional in single and multi modes, currently v1.0.2.9001.

3.10.1 Module origin and single/multi mode

- Moved burn post-processing out of the main LandWeb repo into a self-contained module (2023-09); reached v1.0.0 with an `scfm` update and terra adoption (2024-06).
- Single/multi refactor: single mode saves the inputs multi needs as its own outputs (avoiding unreliable `simList` loads), incorporates `timeSinceFire`, adds `registerOutputs()`, `qs`, terra vectors, and `tidyterra+ggplot2` plotting in place of `rasterVis` (2026-02).

3.10.2 `start(sim)` integration and `summaryPeriod` robustness

- Offset `analysesOutputsTimes` and the scheduled `save_single` events by `start(sim)`; added an init-time guard that `summaryPeriod` falls within `start(sim)/end(sim)` (2026-03).

3.10.3 `fireSense` interoperability

- Accepts differently-named `fireSense` objects (`flammableRTM` for `flammableMap`, `nonForest_timeSinceDisturbance` for `rstTimeSinceFire`); declares `loadOrder = list(after = c("fireSense", "LandMine", "scfmSpread"))` (2026-03).

3.10.4 NFDB retrieval workaround and cumulative burn map

- Introduced the historical cumulative (mean-annual) burn map from NFDB fire polygons (2023).
- Replaced the failing `prepInputs()` NFDB fetch with a manual `download.file` + `archive::archive_extract`, dropping invalid geometries via `terra::is.valid` (avoiding slow `makeValid`) and combining via `tidyterra::bind_spat_rows`; added `archive/purrr` to `reqdPkgs` (2026-03).

3.11 LandWeb_summary

The v2 post-processing/reporting module that aggregated per-rep LandR outputs into leading-species, large-patch, and stand-age products. **In v3 it is superseded by `NRV_summary` + `nrvtools`**, which fold in-simulation per-rep saves and post-hoc aggregation into one two-phase workflow.

3.11.1 Reporting-polygon and tiling refactor

- Moved from a monolithic `map` object toward a `reportingPolygons` list (WIP, 2025-08); removed the `.makeTiles` parameter, dropping the old tiling path (2026-01).

3.11.2 Post-processing split and stand age

- Split post-processing by `type` so stages can run across multiple machines; tuned RAM estimates and switched core detection to `parallelly::availableCores()` (2025).
- Reworked stand age to derive from cohort data rather than time-since-fire, adding a `standAgeMapFromCohorts` param with TSF fallback (2023-2024).

3.11.3 Retirement in v3

- Its leading-type/large-patch/stand-age summarization role is taken over by `NRV_summary` (single- then multi-mode aggregation) and `nrvtools`.

3.12 LandWeb_output

The v2 module that wrote per-simulation LandR output maps (notably the stand-age map) to disk for downstream reporting. **In v3 it is retired**, superseded by `NRV_summary`'s in-simulation per-rep saves plus `nrvtools`.

3.12.1 terra adoption and stand-age from cohorts

- Switched from raster to terra (2023-09); produced `standAgeMap` from `cohortData` (biomass-weighted ages), bumping the LandR version requirement (2023-10).

3.12.2 Retirement in v3

- Its output-writing role is absorbed by `NRV_summary`'s in-sim per-rep saves; `nrvttools` handles downstream products.

3.13 timeSinceFire

The v2 module that computed and wrote the time-since-fire (TSF) raster used by stand-age and reporting. **In v3 it is retired:** TSF is now derived post-hoc, and its role is superseded by `NRV_summary` + `nrvttools`.

3.13.1 terra/raster migration and paths

- Reverted to raster from terra to fix a `compareGeom()` `ext/extent` mismatch (2023-09), then later re-migrated to terra with refreshed docs (2025-10); removed a duplicate metadata block left by a merge (2026-02).
- Switched to `inputPath()` over `dataPath()`; removed a stray `browser()` (2023-09).

3.13.2 Retirement in v3

- TSF is now computed post-hoc rather than as a standalone module; `NRV_summary` + `nrvttools` cover the downstream need.

3.14 HSI_Caribou_MB

Manitoba caribou habitat-suitability-index module; only light maintenance since 2023, and retained (not retired) in v3.

3.14.1 Maintenance

- Fixed a partial-match warning by renaming the `rasterListByPoly()` argument `poly` to `polys` (2024-02).
- Manual/gitignore housekeeping for the PDF manual build (2023).

4 Owned R packages

Much of v3 lives in packages the project owns and CI-tests, several created specifically for the refactor. These absorb the generic glue (so the modules stay thin) and the LandWeb-specific logic.

4.1 SpaDES.targets

A new-for-v3 package that is the SpaDES-to-targets firewall and glue: it runs a `simInitAndSpades()` stage in-process as a `targets` stage and, instead of serializing the whole `simList`, returns a manifest of the files the stage saved (discovered from `outputs(sim)`) as `format = "file"` targets, plus a hard options “firewall” that disables the destructive/non-headless `reproducible/SpaDES.core` behaviours. Entire history is June 2026.

4.1.1 The options firewall

- `spades_safe_options()` / `with_spades_safe_options()`: the single adapter that makes `targets` the sole cache and disables destructive/non-headless-safe behaviours
 - `reproducible.useCache/useMemoise/useCloud = FALSE`, `spades.saveSimOnExit = FALSE` (never serialize a `simList`), `spades.browserOnError = FALSE` and `reproducible.interactiveOnDownloadFail = FALSE` (guard a worker hanging on `browser()/a download prompt`), and `recoveryMode/futureEvents/allowInitDuringSimInit` off.
- A `strict = TRUE` switch turns development diagnostics back on; production default leaves them off for speed/memory. A cross-version rename shim sets both retired and current option names (e.g. `reproducible.rasterRead = "terra::rast"`); audited against `SpaDES.core 3.1.2.9016` / `reproducible 3.1.1.9062`.

4.1.2 The `outputs(sim)` manifest API

- `extract_outputs()` reads the files a stage actually saved after a run and returns a `manifest` plus a `files` vector, capturing per-timestep saves, module `registerOutputs()` dumps, and `Plots()` figures with no a-priori declaration (replaced the older `extract_components()`, a breaking change) (2026-06-26).
- `outputs_spec()` builds a `simInit(outputs=)` table grouped by save function and expanded over `saveTime` (later added `qs/csv` groups); emits only `fun/package` strings so it adds no dependencies.

- `sim_inputs()` builds a `simInit(inputs=)` table from an upstream manifest so a downstream stage reloads file-backed outputs; `sim_objects()` loads a manifest's outputs into memory for `simInit(objects=)` (spatial handoffs a downstream module touches in `.inputObjects()`).

4.1.3 Target factory, worker, and per-run scratch isolation

- `tar_simspades()` target factory + `run_simspades()` worker run one stage in-process with no `simList` serialized; the factory emits the manifest target plus a `<name>_files` format = "file" companion tracking every saved file, with a `loadOrder` pass-through for ambiguous auto-inference.
- Per-run scratch isolation: each stage runs in a unique subdir under `paths$scratchPath` (concurrent runs do not collide); `out_dir` cleared at start; a success removes the scratch dir, an R-level failure keeps it renamed `*.FAILED`, and stale dirs older than `scratch_retain_days` (default 7) are swept on the next run.

4.1.4 Path portability and spatial/provenance helpers

- Manifest paths stay project-relative even when a stage writes through a symlinked output dir to NFS (re-relativized to the deepest shared component rather than escaping to `../../../../mnt/...`).
- `write_spatial()/read_spatial()/is_spatial()` terra-first spatial file-target helpers (with an `sf`-to-`SpatVector` bridge for not-yet-converted modules); `provenance_manifest()` records R version/platform, `renv.lock` md5, `modules/` pinned commits, and per-output md5 digests.

4.2 spatialutils

A new-for-v3 package collecting generic, project-agnostic `sf/terra` spatial helpers: a robust vector-prep pipeline, polygon cleaning/overlay operations, and small convenience wrappers. It underpins the v3 reporting/preamble geometry handling.

4.2.1 Vector prep pipeline

- `prep_vector()` (2026-06-25) crops, masks, and reprojects a vector to a study area while guaranteeing valid topology: source-CRS pre-crop, `terra::makeValid()` repair, optional `mask`, and a configurable `buffer`.

4.2.2 Polygon cleaning and overlay operations

- `eliminate_slivers()` (alias `desliver()`) merges sliver polygons into the neighbor with the longest shared border (2026-06).
- `st_union_analysis()` emulates `arcpy Union_analysis` with multi-attribute transfer and a `union_by` column, later sped up and refactored to use `st_erase()` (2026-01); additional ops `intersect_clean()`, `outerBuffer()`, `perimeter()`.

4.2.3 Generic `sf`/terra helpers, licensing, modernization

- `create_bbox()`, `calc_centroid()`, `extract_coords()`, `save_gpkg()`, `spatial_join_intersects()`, and `prep_agg_lonlat_raster()` added as one batch (2026-06-17).
- Dropped the `ngeo` dependency for `sf::st_exterior_ring()`; relicensed to Apache-2.0; bumped to R >= 4.3; added NEWS and testthat-3 snapshot tests.

4.3 nrvtools

A new-for-v3 package providing reusable analytics for Natural Range of Variability (NRV) simulation outputs: patch- and landscape-level metrics, seral-stage mapping, configurable metric sets, baked-in Canadian ecological polygons, and `ggplot2` summary plots. It is the shared engine behind the v3 NRV_summary/burnSummaries reporting.

4.3.1 Patch and landscape metrics engine

- Core exports `calculatePatchMetrics()/calculatePatchMetricsSeral()` (v0.0.2, 2024-05) and `calculateLandscapeMetrics()`, plus lower-level `patchStats()/patchAreas()/summarizeP` helpers.
- Metrics via `landscapemetrics` with per-polygon extraction through `exactextractr::exact_extract()`; parallelized over polygons with `future.apply::future_lapply()` (hardened RNG/globals).

4.3.2 Seral-stage mapping and configurable metric sets

- `seralStageMapGeneratorBC()/writeSeralStageMapBC()` build BC seral-stage maps from current-condition, PGM, and NDT/BEC inputs (iterative factor-level/NDT4 fixes through 2024-06).
- `default_landscape_metrics()/default_patch_metrics()/default_patch_metrics_seral()` supply curated lists; all calculators accept a user `funList` override.

4.3.3 Baked-in ecological polygon data

- Bundles `ecozones` (25), `ecoprovinces` (68), `ecoregions` (218), and `ecodistricts` (1025) as `sf` datasets sourced from `open.canada.ca` (2026-03-19), generated via `data-raw/eco.R` with `LazyData` gzip compression.

4.3.4 Plotting and modernization

- `plot_over_time()/plot_over_time_by_class()/plot_by_class()` produce `ggplot2` summaries with `ggforce::facet_wrap_paginate()` pagination.
- Migrated from `raster` to `terra`; required R ≥ 4.1 for the native pipe; added `exactextractr/tidyterra/qs2` deps (moved `qs/SpaDES.tools` to `Suggests`); adopted the `air` formatter, a GHA R-CMD-check, and `testthat-3` tests.

4.4 workflowtools

A new-for-v3 grab-bag of project-workflow helpers for large `SpaDES/targets` projects: idempotent “once” download/extract helpers, Google Drive and Microsoft Teams/SharePoint sync, an input-data manifest with one-way BibTeX export, and `git+renv` reproducibility/provenance capture. First committed 2023-09, reaching v0.0.13 (2026-06-09).

4.4.1 Idempotent “once” fetch/extract helpers

- `download_once()` and `drive_download_once()` skip the transfer when the destination exists; `archive_extract_once()` skips re-extraction when the target files are present (selection by filename or numeric index, plus `force`) (2026-03).

4.4.2 Google Drive and Microsoft Teams/SharePoint sync

- `drive_download_folder()/drive_upload_folder()` recursively mirror Drive folders in batches (with an HTTP/1.1 hardening helper `.use_http11()` and `with_retry()` backoff).
- New `teams_*()` family mirroring the Drive helpers against a Teams channel’s SharePoint library: `teams_connect()`, `teams_download_folder()` (recursive walk, `exclude` list), `teams_upload_files()`, `teams_ensure_folder()` (2026-06).

4.4.3 Input-data manifest + BibTeX export

- Manifest API tracking input datasets in `data/input_manifest.json`: `register_input()` (idempotent upsert keyed by id), `read_input_manifest()/write_input_manifest()`, `input_manifest_record()/validate_manifest_record()`, `input_manifest_schema()`; the format carries a JSON-LD @context (2026-06).
- `extract_metadata()` S3 generic turns a source into a record (URL + HEAD Last-Modified probe; `bcdc_record`). `sync_manifest_to_bibtex()` is a conservative one-way exporter to a sidecar `.bib`, deduplicated against `references.bib`.

4.4.4 Reproducibility / provenance capture

- The `info_*`() introspection family – `info_git()`, `info_submodules()`, `info_session()`, `info_spatial_libs()`, and the aggregator `info_project()` – plus `project_path()/project_name()` (snake_case migration, 2026-06).
- `reproducibility_receipt()` wraps these into a git-repo + R-session provenance block for Rmd/Quarto; `packages_from_snapshot()/description()` derive project package metadata from an `renv/Require` snapshot.

4.5 crew.ssh

A new-for-v3, experimental, currently private `crew` launcher plugin that runs `targets` workers on remote machines over SSH, to distribute a pipeline across several computers on a trusted local network. Workers are placed across nodes by per-node capacity weight so heterogeneous machines share one controller – a lightweight counterpart to `crew.cluster`. All history is June 2026 (v0.0.4).

4.5.1 SSH launcher, controller, and node model

- `crew_class_launcher_ssh` (R6) places workers across nodes in proportion to per-node capacities; `crew_controller_ssh()` creates the SSH controller; `crew_ssh_node()` describes one machine (host, capacity, optional `rscript/projdir/ssh_options`) for heterogeneous clusters.

4.5.2 Connectivity and robustness

- `request_tty = TRUE` forces `ssh -tt` so the remote R exits when the local ssh is killed; `tunnel = TRUE` dials the dispatcher back through an SSH reverse tunnel so no inbound port need be open (2026-06).

- `crew_ssh_check()` preflight-checks each node (connectivity, project dir, resolved Rscript/R version, package availability) with an S3 print method.

4.5.3 Live monitoring and node provisioning

- `crew_ssh_monitor()` opens a live CPU/RAM dashboard (a `miniUI` gadget) polling each node over SSH on a timer; reworked to poll with concurrent `processx` children (forking inside a running Shiny gadget was unsafe), degrading a failed poll to “unreachable” (2026-06).
- Bundles a project-agnostic `inst/templates/sync-nodes.R`: provisions nodes in parallel, groups by OS codename so the first node of each group warms a shared `renv` cache before the rest link it, and runs `git submodule sync` before update so a repointed fork URL is picked up (2026-06).

4.6 LandWebUtils

Pre-existing LandWeb-specific helper package, **heavily refactored for v3**: migrated off `sp/raster` to `sf/terra`, slimmed its dependency stack (notably dropping `LandR`), and gained a reporting-polygon/study-area builder plus test and formatting infrastructure.

4.6.1 Consolidating LandWeb and LandMine logic

- Absorbed LandWeb-specific functions formerly in `LandR` (2023-01) and the LandMine module functions (2023-09): `landmine_burn1()`, the `landmine_optim_*` cluster/fit family, and diagnostic plots `landmine_plot_FRI/_LTHFC/_areaBurnedOverTime`; extended `landmine_burn1()` for “burny” scenarios and switched optimization to `parallelly` clusters (2024-2025).

4.6.2 sf/terra migration

- Replaced `sp/raster` with `sf/terra` across the package (2025-08), touching `polygonClean()/cleanLandWebStudyArea()`, `LargePatches()`, `LeadingVegTypeByAgeClass()`, `rasterListByPoly()`, and the LandMine helpers; `targetCRS/LandWebCRS` aligned to `SCANFI` (2025-09).

4.6.3 Reporting-polygon and study-area builder

- New `buildReportingPolygons()`, `reportingPolygonLayers()`, `joinReportingPolygons()`, plus prep helpers `prepFMAs()`/`prepFMUs()`/`prepANSRs()` and `prepStudyArea()`/`prepTestStudyArea()`; study-area lookups `LandWebStudyAreas`, `studyAreaProv()`/`studyAreaIn()`, `extractFMA()`/`extractFMU()` (with a name-fallback).
- Geometry prep delegated to `spatialutils::prep_vector()`; inputs fetched via `workflowtools` idempotent `*_once` helpers.

4.6.4 Dependency slimming, modernization, testing

- Dropped `LandR` from Imports/Remotes, vendoring `equivalentName()`/`equivalentNameColumn()`/`equivalentNameColumn()` (GPL-3) into `R/equivalentName.R` (2026-06); replaced `SDMTools` with `landscapemetrics` (2025-10). (`map` and `pemisc` remain in use.)
- Bumped to `R >= 4.4`; adopted the native pipe (`|>`) and `air` formatting; added `testthat-3` snapshot tests and refreshed GHA workflows.

5 Framework changes and the dependency firewall

The recurring difficulty across v3 was not the ecological code but upstream framework practice. `SpaDES.core` (Chubaty and McIntire 2025) and `reproducible` (McIntire and Chubaty 2024) ship large, non-atomic commits (fixes bundled with regressions) and periodically change user-facing APIs; because fixes are not backported, the only way to obtain one is the unstable development branch. `LandWeb v3` insulates itself with a layered strategy:

- **Pin exact revisions.** Modules and framework packages are pinned to specific SHAs (the `Biomass_*` modules to FOR-CAST forks' `main`, not upstream `development`), so an upstream change cannot silently alter a run.
- **One anti-corruption layer.** All `SpaDES.core` / `reproducible` touch-points funnel through the owned `SpaDES.targets` package. Its `spades_safe_options()` asserts a hardened set of ~14 options (disabling destructive/unsafe development defaults, and `reproducible.useCache = FALSE` so `targets` owns all caching). An upstream break becomes one adapter fix, not many.
- **Refactor what we own; wrap the rest.** Owned modules (`LandWeb_preamble`, `burnSummaries`, `NRV_summary`) are refactored directly; the shared `Biomass_*` modules + `LandMine` are wrapped and fork-pinned.
- **Discrete object hand-off.** Events increasingly pass plain objects rather than serialized `simLists`, so `targets` can drive the modules directly.

Concrete upstream issues encountered and worked around:

Issue	Symptom	Workaround
SCANFI FAO reproject	full-Canada reproject before crop (many minutes)	crop in source CRS first (seconds)
<code>prepInputs</code> empty <code>cropTo</code>	silently returns the whole raster	guard for empty geometry before crop
<code>overlayLCCs</code> all-NA layer	<code>pixelIndex</code> type mismatch on join	feed a filler layer instead of all-NA
Drive restricted files	content download returns an HTML sign-in page	<code>workflowtools::drive_download_once()</code>
reproducible #447 (WONTFIX)	service-account Drive auth deprecated	direct <code>googledrive::drive_auth()</code>
nested-module <code>reqdPkgs</code>	inner <code>simInit</code> skips loading required pkgs	re-assert <code>loadReqdPkgs = TRUE</code> before nested run

Platform-wise, v3 moved to **R 4.6** (from 4.5), forcing a clang-21 rebuild and a full `renv` relocl, and adopted `qs2`, `cli`, `tidyterra`, `exactextractr`, `landscapemetrics`, and `parallelly`.

6 Module supersession

Three v2 post-processing modules are retired in v3, consolidated into one module plus one CI-tested package:

- `LandWeb_summary` + `timeSinceFire` + `LandWeb_output` are retired into `NRV_summary` (the module) and `nrvttools` (the reusable analytics + plotting package).
- Time-since-fire is derived in post-processing from burn maps rather than tracked by a dedicated simulation module.
- The reporting-polygon build moved out of `LandWeb_preamble` into a data-driven builder (`LandWebUtils::buildReportingPolygons()`), feeding the summaries stage.

The `LandWeb_full` study area is also dropped: v3 simulates and reports per FMA rather than maintaining a single expensive full-extent run.

Appendix: change summary (auto-generated)

Generated from each component's git log at render time (`R/module_changelog.R`); `commits` counts commits since 2023-01-01, the v3 window.

Table 1: SpaDES module submodules: commits since 2023, last update, and pinned revision.

Module	Commits	Last update	Branch	SHA
Biomass_borealDataPrep	377	2026-06-30	main	f570094
Biomass_speciesParameters	225	2026-06-29	main	1e381a7
Biomass_core	192	2026-06-29	main	32a3790
Biomass_speciesData	137	2026-06-29	main	101ad29
Biomass_speciesFactorial	85	2026-06-30	main	01d4559
NRV_summary	76	2026-04-20	development	ca7b444
LandWeb_preamble	59	2026-07-02	development	65dc365
burnSummaries	58	2026-03-27	development	55ebb1f
LandMine	58	2026-07-02	development	d7ccb08
Biomass_regeneration	29	2026-06-02	main	b739da2
LandWeb_summary	27	2026-01-21	main	99e229e
LandWeb_output	15	2026-02-02	development	af78cee
timeSinceFire	12	2026-02-02	development	30d242e
HSI_Caribou_MB	3	2024-02-15	HEAD	1180d61

Table 2: Owned R packages: commits since 2023, last update, and pinned revision.

Package	Commits	Last update	Branch	SHA
LandWebUtils	63	2026-06-25	development	cf29ea0
nrvtools	51	2026-03-26	development	f9665fd
workflowtools	43	2026-06-09	development	561def4
spatialutils	34	2026-06-25	main	a891aef
SpaDES.targets	15	2026-06-30	main	b1f68a5
crew.ssh	10	2026-06-30	main	79c368f

References

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Provenance

- **Source:** the git history of the module submodules (modules/) and owned packages (packages/), plus the refactor design notes (`_tmp_refactor_targets.md`, `_tmp_spades_targets_bchrv_gaps.md`, `_tmp_upstream_issues.md`).
- **Data file:** `outputs/extended_analyses/module_changelog.csv`
- **Method:** the change-summary tables are built by `R/module_changelog.R` (target `module_changelog_csv`), which runs `git log` per component at render time; the narrative themes are hand-curated from the same history.

Appendix: Build provenance

This appendix pins the rendered PDF to the code, environment, and revisions that produced it, so a result can be reproduced or a difference attributed to code vs. inputs vs. randomness. It is generated at render time from the `workflowtools info_*`() family.

Build identity. LandWeb repository, branch `targets`, commit `ad683998`, rendered 2026-07-03 10:18 MDT with R version 4.6.1 (2026-06-24) on Ubuntu 26.04 LTS (GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1).

Table 3: SpaDES modules and owned packages at their pipeline-pinned revisions when this report was rendered (`workflowtools::info_submodules()`).

Component	Kind	Pinned SHA
Biomass_borealDataPrep	module	f570094ea9
Biomass_core	module	32a3790870
Biomass_regeneration	module	b739da2d32
Biomass_speciesData	module	101ad29571
Biomass_speciesFactorial	module	01d4559d15
Biomass_speciesParameters	module	1e381a7881
HSI_Caribou_MB	module	1180d61743
LandMine	module	d7ccb083e1
LandWeb_output	module	af78cee648
LandWeb_preamble	module	65dc365f37
LandWeb_summary	module	99e229e3ac
NRV_summary	module	ca7b444fa2
burnSummaries	module	add6cc0bb4
timeSinceFire	module	30d242ea7b
LandWebUtils	package	cf29ea06e4
SpaDES.targets	package	b1f68a5467
crew.ssh	package	79c368f0fa
nrvttools	package	f9665fd7b0
spatialutils	package	a891aef9d5
workflowtools	package	561def4389

The full `sessionInfo()` and the exact `renv.lock` package pins accompany the run; input datasets are recorded in `_input_manifest.json` (report 06).

DRAFT